Econometric Workshop Final Project
Partial Identification of Tutoring Effects

Francesco Dotti Giberti

Abstract

This project studies tutoring effects in a Chilean university leveling program using partial identification and strong-ignorability benchmarks. Starting from the Manski worst-case bounds, which require only bounded outcome support, the analysis adds Monotone Treatment Response (MTR) and Monotone Treatment Selection (MTS) restrictions to progressively tighten the identified intervals. Under bounded support alone, the sign of the Average Treatment Effect (ATE) is not established for any threshold or subgroup in either semester: the grade ATE interval spans approximately $[-2.7, 3.6]$ at the program's operational threshold $k = 10$, and pass-rate intervals are similarly wide. Under MTR the grade lower bound shifts to zero, halving the interval width, but at the cost of asserting that tutoring cannot harm any student. The results illustrate Manski's Law of Decreasing Credibility: stronger assumptions yield sharper conclusions but reduce the credibility of the inference.

1 Introduction

The project analyzes a Chilean university tutoring program with more than 3800 students over two semesters. The empirical strategy is organized around two questions:

1. In semester 1, for BAP students, which tutoring intensity threshold k gives the strongest effect on grades and approval rates?
2. In semester 2, for BAP and VAI students, is it better to continue tutoring or opt out?

2 Data Processing

The dataset was processed to ensure consistent variable definitions across both semesters and all subgroups. The core variables selected are: academic outcomes, tutoring attendance, baseline admission scores, and subgroup indicators. Binary indicators such as BAP and VAI were treated consistently as group labels, while semester-specific treatment variables were constructed separately for each question.

For Question 1, treatment intensity was defined through threshold rules of the form $Z_k = \mathbf{1}[\texttt{Tut\_20161} \geq k]$. For Question 2, the continuation decision in semester 2 was defined from attendance information as Continue versus Opt-out, where $\texttt{Continue}_2 = \mathbf{1}[\texttt{Tut\_20162} \geq 1]$ and a missing value for `Tut_20162` was treated as zero attendance. Missing outcome values were preserved rather than imputed, since the identification strategy explicitly incorporates the uncertainty arising from incomplete records.

3 Identification Strategy

3.1 The Identification Problem

The parameter of interest is the Average Treatment Effect

$$\tau \;=\; \mathbb{E}[Y_i(1) - Y_i(0)], \tag{1}$$

where $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes of student i under treatment ($D_i = 1$) and control ($D_i = 0$) respectively. The ATE is not point-identified from observational data. This follows directly from the Fundamental Problem of Causal Inference: for any student i we observe at most one of the two potential outcomes. A student who attended at least k sessions ($D_i = 1$) has observed outcome $Y_i = Y_i(1)$, while her counterfactual $Y_i(0)$ is never observed; and symmetrically for non-participants.

The data generating process reveals only the joint distribution of (Y_i, D_i), from which we can identify $\mathbb{E}[Y \mid D = 1]$ and $\mathbb{E}[Y \mid D = 0]$. These are conditional means on factual arms, not population potential-outcome means. By the Law of Total Probability, the potential-outcome means decompose as

$$\mathbb{E}[Y_i(1)] = \mathbb{E}[Y \mid D = 1]\, P_1 + \underbrace{\mathbb{E}[Y_i(1) \mid D = 0]}_{\text{unidentified}}\, P_0, \tag{2}$$

$$\mathbb{E}[Y_i(0)] = \mathbb{E}[Y \mid D = 0]\, P_0 + \underbrace{\mathbb{E}[Y_i(0) \mid D = 1]}_{\text{unidentified}}\, P_1, \tag{3}$$

where $P_1 = P(D = 1)$ and $P_0 = 1 - P_1$. The first terms on the right-hand sides are identified from the data. The second terms, $\mathbb{E}[Y_i(1) \mid D = 0]$, the mean outcome that non-participants *would* achieve under treatment, and $\mathbb{E}[Y_i(0) \mid D = 1]$, the mean outcome that participants *would* achieve without treatment, are counterfactuals and are therefore fundamentally unidentified by any observational study.

This gap cannot be closed by collecting more data under the same design. Any point estimate of τ requires an untestable identifying assumption. The approach taken here is to make those assumptions explicit and to report how conclusions change as they are varied.

3.2 Manski Bounds via the Law of Total Probability

Without any restriction beyond bounded support $Y_i(t) \in [y_{\min}, y_{\max}]$, the unidentified counterfactual means are bounded by the support limits.

Proposition 1 (Manski sharp bounds)**.** *Under the support restriction $Y_i(t) \in [y_{\min}, y_{\max}]$ and no further assumptions, the sharp identification region for τ is the closed interval $[\tau^L, \tau^U]$ where*

$$\tau^L = \big[\mathbb{E}[Y \mid D = 1]\, P_1 + y_{\min}\, P_0\big] - \big[\mathbb{E}[Y \mid D = 0]\, P_0 + y_{\max}\, P_1\big], \tag{4}$$

$$\tau^U = \big[\mathbb{E}[Y \mid D = 1]\, P_1 + y_{\max}\, P_0\big] - \big[\mathbb{E}[Y \mid D = 0]\, P_0 + y_{\min}\, P_1\big]. \tag{5}$$

Remark 1 (Width and missing data). Simplifying $\tau^U - \tau^L$ gives

$$\tau^U - \tau^L \;=\; (y_{\max} - y_{\min})(P_0 + P_1) \;=\; y_{\max} - y_{\min}. \tag{6}$$

The width equals the full outcome range, independent of sample size or treatment shares. For grades ($y_{\min} = 1$, $y_{\max} = 7$) the theoretical maximum is 6; for approval rates ($y_{\min} = 0$, $y_{\max} = 1$) it is 1. In the implementation, rows with missing outcomes are dropped, so the computed bound is on the observed-outcome subset; the resulting width is the $(y_{\max} - y_{\min})$ value of equation (6) applied to that subset, which is slightly below the theoretical maximum once finite-sample factual means are plugged in.

3.3 Assumption Tightening: MTR and MTS

The Manski bounds are wide because no restriction is placed on treatment response or selection. Two distributional assumptions narrow the interval while remaining empirically motivated.

The two restrictions introduced below, Monotone Treatment Response and Monotone Treatment Selection, are due to [4] and fit naturally inside the lecture framework. Both can be cast as stochastic-dominance restrictions on counterfactual distributions in the sense of §3.3 of the lecture notes: MTR is equivalent to requiring $F_{Y(1)|X}$ to first-order stochastically dominate $F_{Y(0)|X}$ pointwise in i, which is the individual-level analogue of the parameter monotonicity discussed there; MTS is equivalent to requiring $F_{Y(t)|D=1}$ to dominate $F_{Y(t)|D=0}$ in the same sense. They therefore play the same role in tightening the Manski bounds as the stochastic-dominance restrictions of §3.3 and the mean-monotonicity instrument-style restriction of §3.6, applied here to the treatment indicator rather than to an external instrument.

Assumption 1 (Monotone Treatment Response, MTR)**.** For all i: $Y_i(1) \geq Y_i(0)$.

MTR formalizes the prior belief that tutoring cannot harm any student. Under Assumption 1, $\tau = \mathbb{E}[Y_i(1) - Y_i(0)] \geq 0$, so τ^L is replaced by $\max(\tau^L, 0)$. The upper bound is unchanged.

Assumption 2 (Monotone Treatment Selection, MTS)**.** For $t \in \{0, 1\}$: $\mathbb{E}[Y_i(t) \mid D = 1] \geq \mathbb{E}[Y_i(t) \mid D = 0]$.

MTS captures positive selection: students who chose more tutoring have weakly better potential outcomes on average, regardless of the treatment level. Under Assumption 2 the lower bound of $\mathbb{E}[Y_i(1)]$ is tightened: instead of using $y_{\min}$ for the counterfactual component, it uses $\mathbb{E}[Y \mid D = 0]$, i.e. the observed control mean, as the floor. In practice MTS narrows the interval relative to Manski but less aggressively than MTR.

Law of Decreasing Credibility [1]: the credibility of inference decreases with the strength of the assumptions maintained. The assumption ladder Manski → MTS → MTR makes this trade-off explicit: each step tightens the interval but adds a non-refutable restriction on counterfactual outcomes.

3.4 Strong Ignorability Benchmark

Assumption 3 (Strong Ignorability [3])**.** $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$, where X_i denotes observed pre-treatment covariates.

Under Assumption 3 the ATE is point-identified. Two versions are reported. The *unconditional* version sets $X_i = \emptyset$, giving $\tau = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$, a simple difference in observed means. The *conditional* version uses the baseline admission scores

$X_i = (\text{PTJE\_RANKING}, \text{PTJE\_LYC}, \text{PTJE\_MAT}, \text{VAI})$ and is implemented by g-computation (outcome regression): fit $\hat{\mu}_t(x) = \hat{\alpha}_t + \hat{\beta}_t^\top x$ separately on each treatment arm by OLS, then report $\hat{\tau} = N^{-1} \sum_i [\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)]$.

Two properties of Assumption 3 are important for interpretation. First, it is *not empirically refutable* [2]: no test on the observed data (Y, D, X) can falsify it, because the assumption concerns unobservable counterfactuals. Second, in this dataset the unconditional version is clearly implausible: BAP students self-select into tutoring and differ systematically from non-BAP students in baseline scores (correlation between VAI and language score is -0.35; Figure 5). The conditional version is more defensible but still relies on the untestable assumption that the four observed covariates exhaust the relevant sources of selection. Both estimates are reported as a benchmark, not as the main causal conclusion.

4 Descriptive Analysis

The descriptive analysis focuses on the BAP and VAI subgroups, measuring the number of students in each subgroup and outcome distributions across semesters, as these are the key variables for the subsequent causal analysis. The descriptive statistics highlight the extent of missingness in tutoring variables, which motivates the use of partial-identification methods to account for this uncertainty.

Even before looking at the graphs we can see that 89.37% and 89.78% of students have missing tutoring data, meaning they did not attend a single tutoring session, in semester 1 and semester 2 respectively. The number of students that are considered actively engaged by the Official Leveling Program guidelines (attended 10 or more sessions) is consequently even lower. This is a crucial point to keep in mind when interpreting the results, as it suggests that the observed tutoring effects may be driven by a small subset of students who attended sessions, and that missing data could bias naive estimates if not properly accounted for, motivating the partial-identification approach of Section 3.

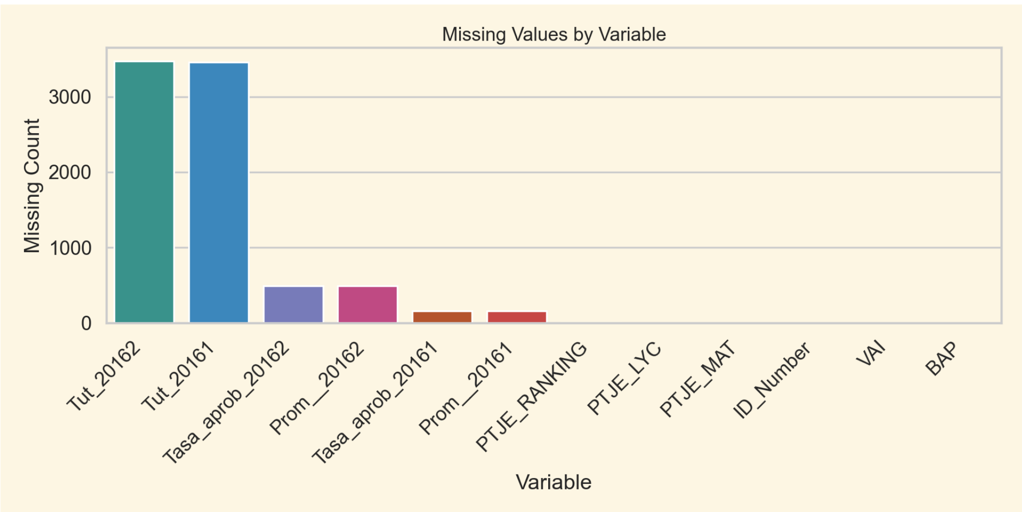

Figure 1: Missing value counts by variable. Tutoring attendance records (`Tut_20161`, `Tut_20162`) account for the vast majority of missing observations (≈3,500 of ≈3,800 students), while outcome variables have far fewer gaps.

Looking at Figure 2 we can see, in the first row, the overall distribution for

PTJE_RANKING, PTJE_LYC and PTJE_MATH. The vertical dashed lines in PTJE_LYC and PTJE_MATH represent the standardized national mean for these exams, showing that students in this sample have mean scores above the national average, with a mean around 600 for both subjects. The first two graphs in the second row show the number of tutoring sessions taken among students who attended at least one session (with mean around 14 and 13 for semesters 1 and 2 respectively). The remaining graphs show approval rates and average grades across the two semesters, with better results in the first semester.

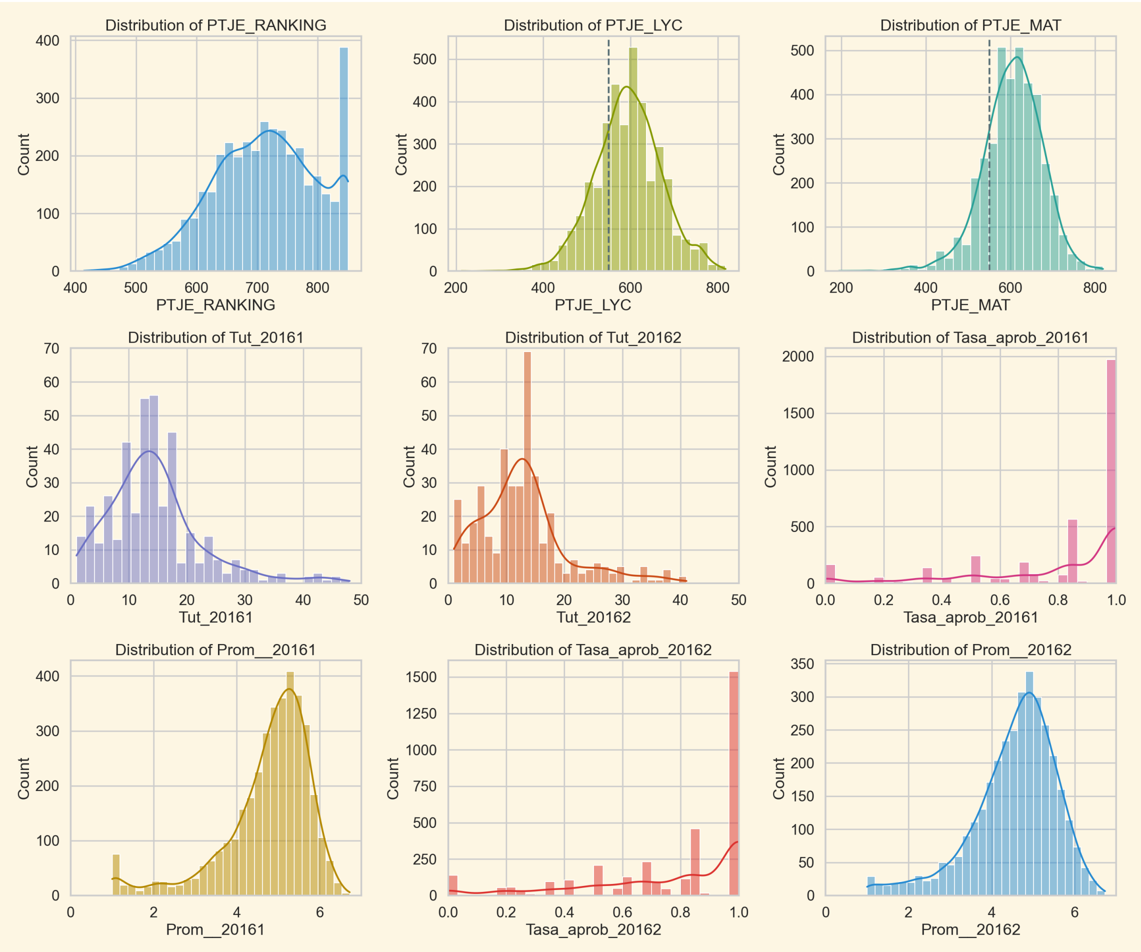

Figure 2: Distributions of key numeric variables.

Figure 3 plots the conditional distributions of approval rates and average grades for BAP vs. non-BAP students. The most important observation is that BAP students appear to perform worse relative to students who did not attend any tutoring sessions. This is counterintuitive only as long as we do not consider that students in the inclusive access program (VAI) are those most incentivized to go to tutoring sessions: they are disadvantaged for many reasons and already constitute a group that we would expect to have lower average performance (as can also be seen in Figure 4). Therefore, looking at conditional distributions is not enough to draw any causal conclusions, and careful interpretation of results is required.

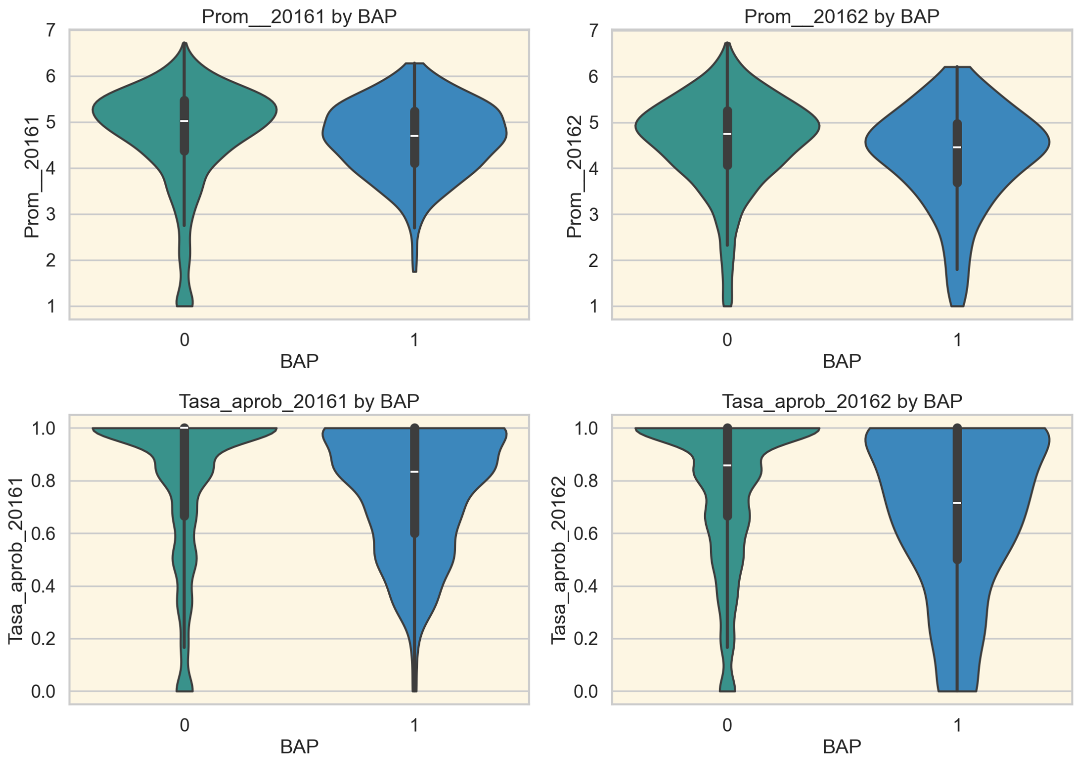

Figure 3: Violin outcome distributions by BAP status across semesters.

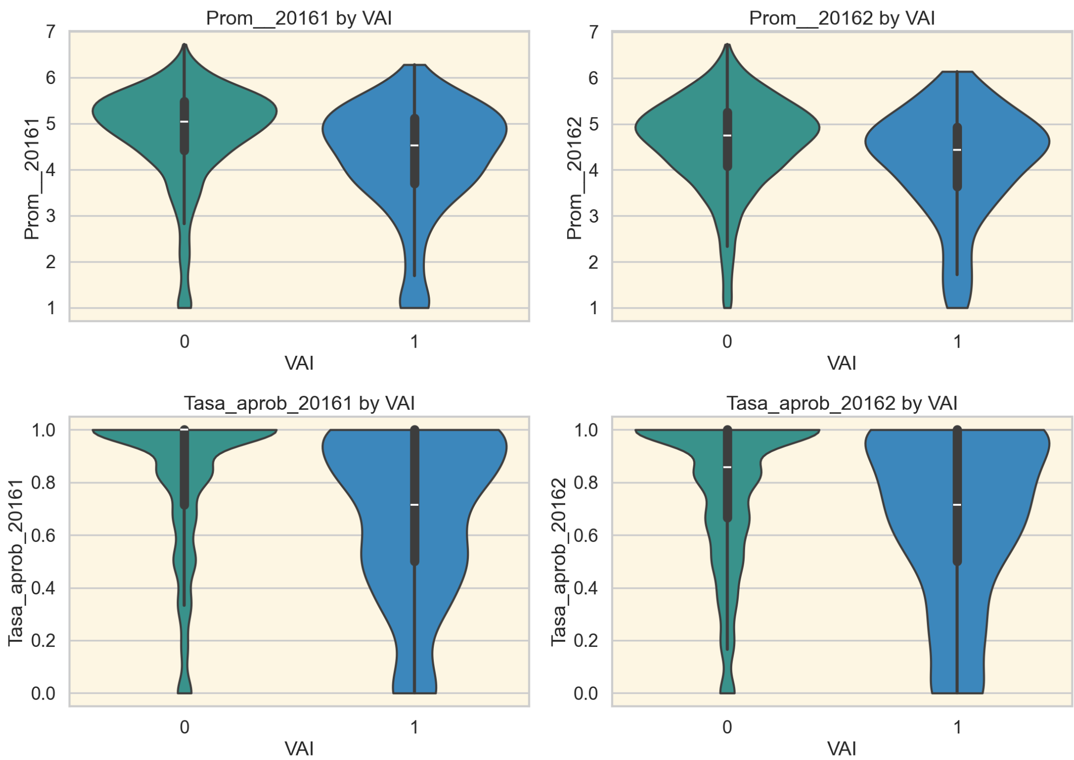

Figure 4: Violin outcome distributions by VAI status across semesters.

In Figure 5 there is a correlation matrix with all numerical variables. The highest correlation (excluding the within-semester grade and approval rate pair) is between VAI and BAP (0.89). As expected, among the lowest correlations are those between the mathematics and language scores and the VAI indicator (-0.35 and -0.32 respectively).

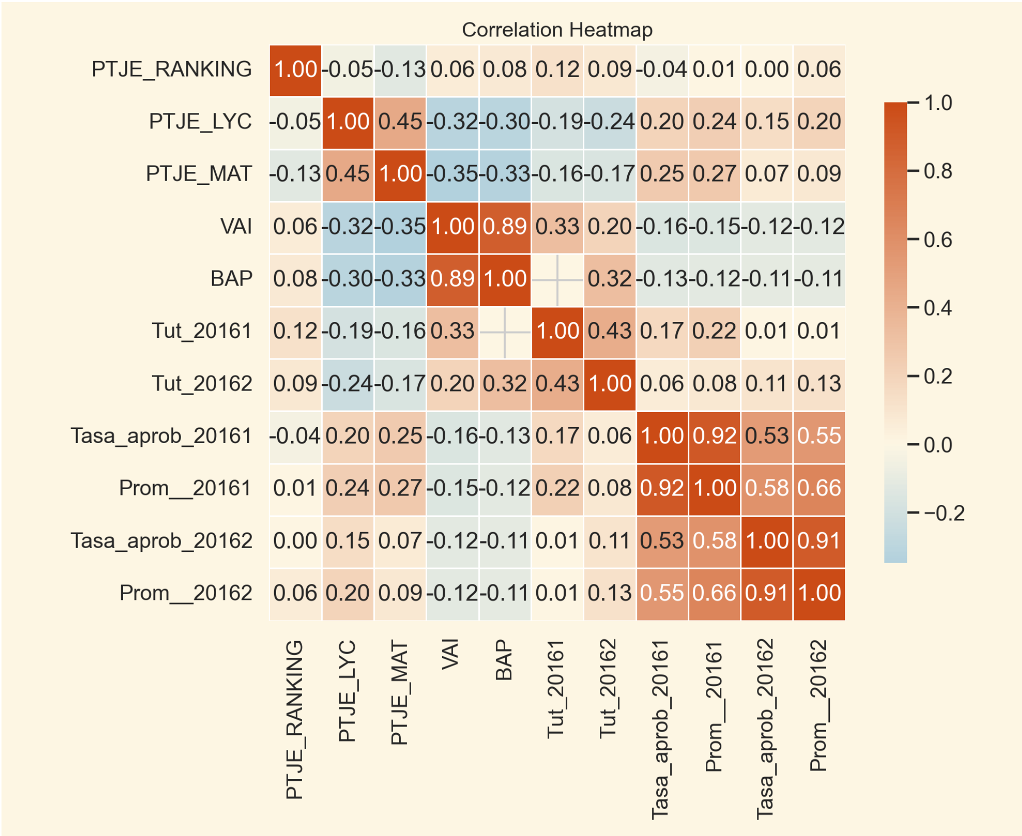

Figure 5: Correlation heatmap for numeric variables.

5 K-Threshold Analysis

For Q1, treatment is defined as $Z_k = \mathbf{1}[\texttt{Tut\_20161} \geq k]$. The main question is how the estimated effect varies with the tutoring threshold.

The first step is to estimate the partial-identification intervals for the average grade and approval rates for each threshold k. The results are presented in Figures 6 and 7. The estimated ATE interval is highly sensitive to how the BAP participation threshold is defined. No robustly positive effect is identified across any threshold: the lower bound remains negative for all k under the Manski assumption.

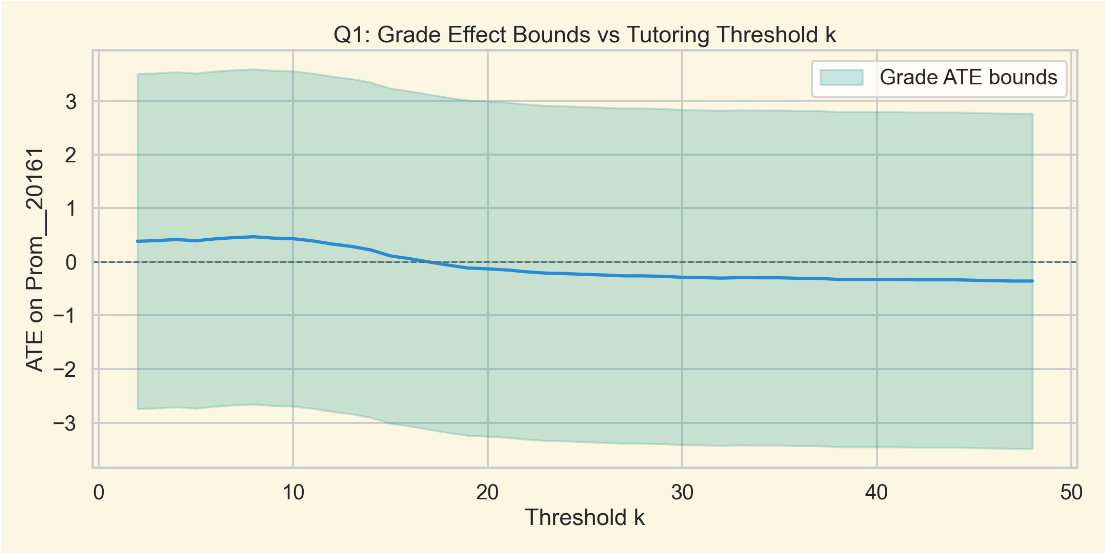

Figure 6: Grade bounds and midpoint by threshold k.

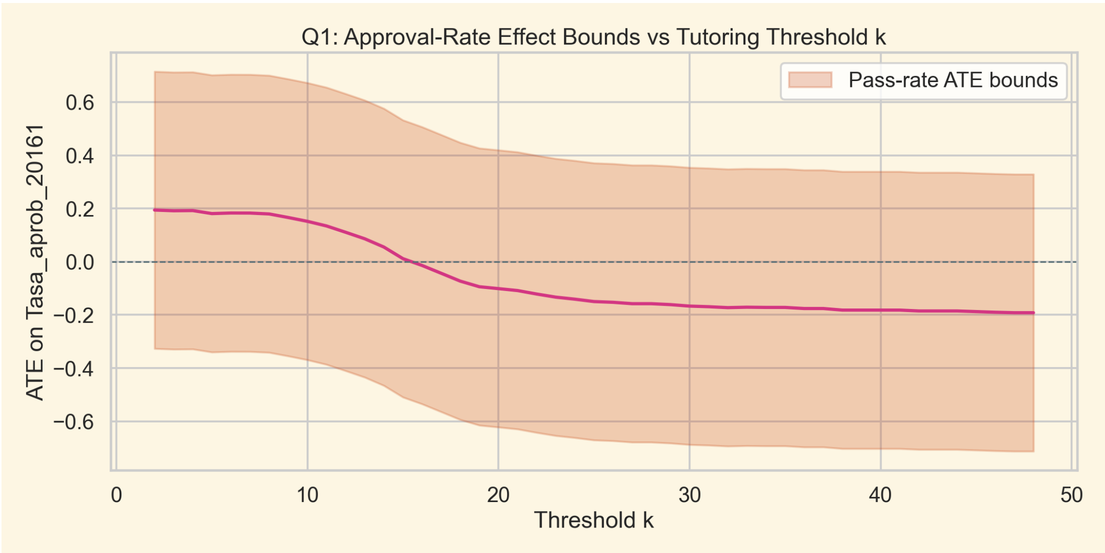

Figure 7: Approval-rate bounds and midpoint by threshold k.

5.1 Assumption Ladder

The analysis extends the Manski bounds with an assumption ladder incorporating MTR and MTS restrictions, together with bootstrap stability analysis.

Figure 8 shows the midpoint of the grade ATE bounds under three partial-identification assumptions as the tutoring threshold k varies from 1 to 50. Under the Manski assumption, which imposes no restrictions on treatment response, the midpoint starts near zero and drifts slightly negative as k increases. Under MTR (Assumption 1), which requires that tutoring cannot harm any student, the midpoint is persistently positive, stabilizing around +1.4 to +1.7 across all values of k. Under MTS (Assumption 2), which accounts for the fact that higher-ability students may self-select into tutoring, the midpoint stays close to zero with a slight positive tendency. The stark divergence across assumptions reveals that the estimated effect of tutoring on grades is highly sensitive to the identifying restriction chosen, and that prior beliefs about selection and treatment response play a decisive role in the conclusion reached.

Figures 9 and 10 compare, for grades and approval rates respectively, the partial-identification intervals under Manski, MTR, and MTS with the strong-ignorability point estimates as k varies from 1 to 50. In the grade panel, the Manski bounds are wide and symmetric around zero, spanning roughly $[-3, +3.5]$, so the sign of the effect cannot be determined from the data alone; under MTR the entire interval lies above zero for all k; and the MTS bounds remain centred around zero but retain an upper endpoint close to $+3$, indicating substantial room for positive effects even under more conservative selection assumptions. In the pass-rate panel, a similar pattern emerges at a smaller scale. The strong-ignorability estimates stay moderately positive across thresholds in both outcomes; the conditional (covariate-adjusted) version is uniformly larger than the unconditional one, reflecting that lower-baseline students are over-represented among tutoring attendees, so the raw difference in means understates the conditional effect.

Taken together, the three figures suggest that a positive effect is plausible and consistent with monotonicity assumptions, but cannot be established conclusively without stronger identifying restrictions.

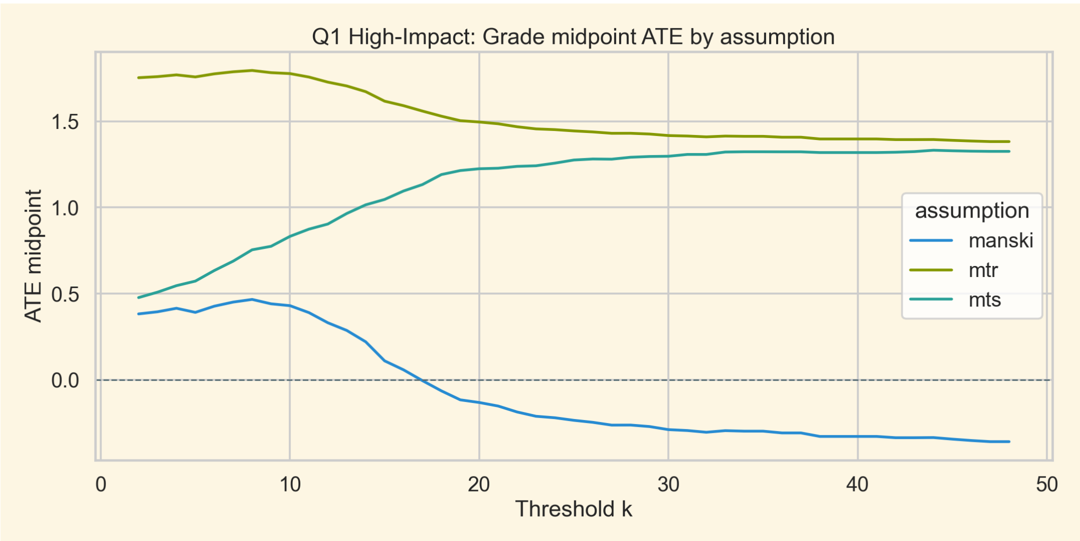

Figure 8: Grade ATE midpoint by assumption as k varies.

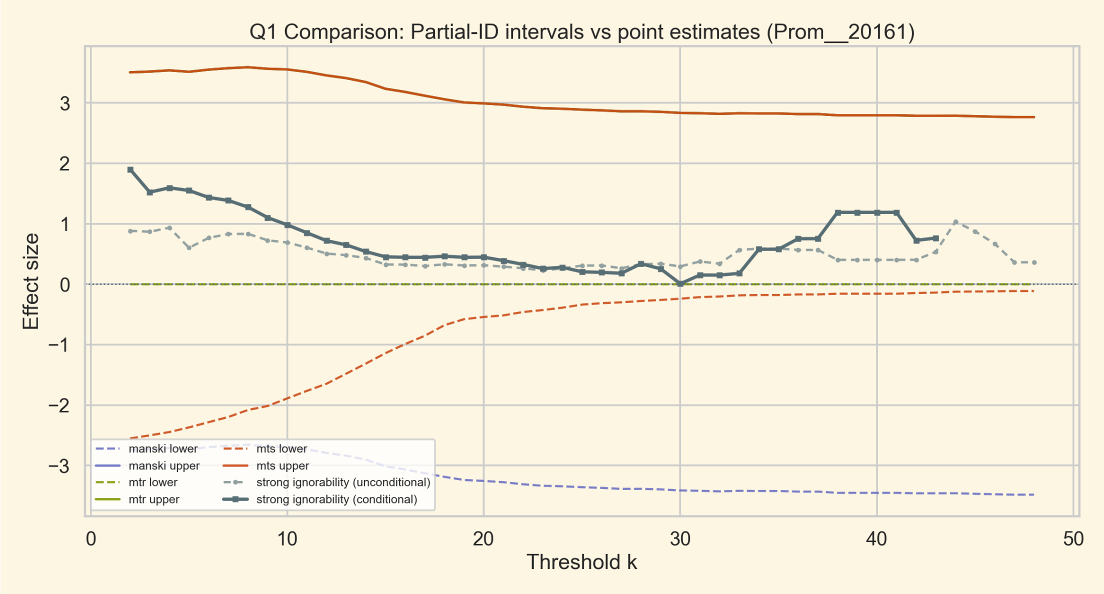

Figure 9: Q1: partial-identification intervals versus strong-ignorability point estimates (grades). The strong-ignorability curve is shown both unconditionally (light dashed) and after covariate adjustment on PTJE_RANKING, PTJE_LYC, PTJE_MAT and VAI (solid). Manski upper bound coincides with MTR upper bound.

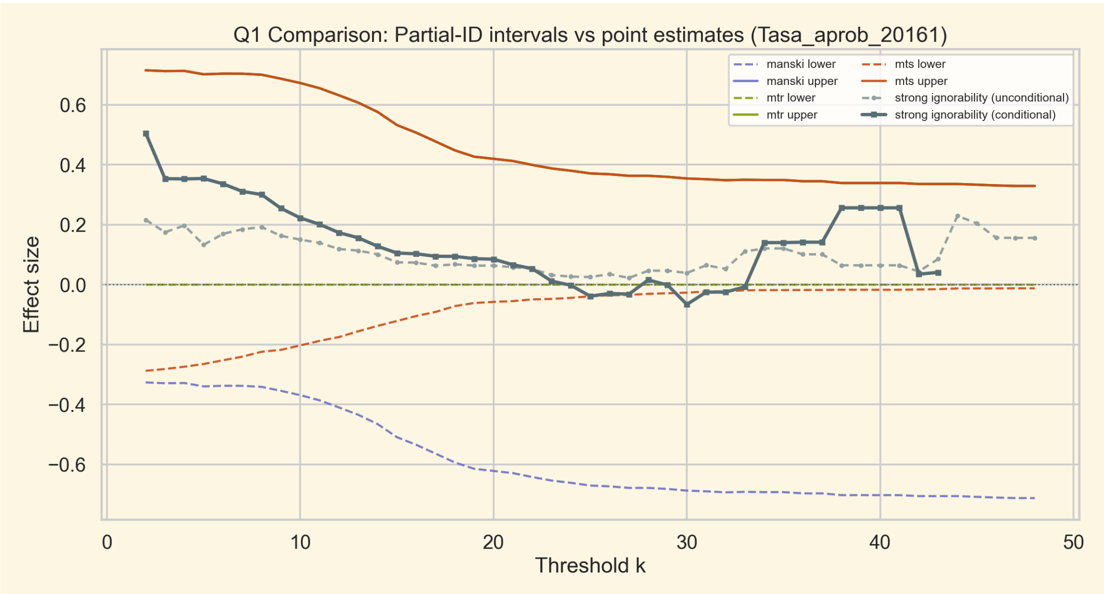

Figure 10: Q1: partial-identification intervals versus strong-ignorability point estimates (approval rate). The strong-ignorability curve is shown both unconditionally and after covariate adjustment. Manski upper bound coincides with MTR upper bound.

5.2 Threshold Recommendation

Table 1 reports the Manski bounds at three decision-relevant thresholds.

Table 1: Manski ATE bounds at selected tutoring thresholds (BAP students, Semester 1). Bracketed values are 5th–95th percentile bootstrap CIs (120 resamples) on each endpoint.

	Average grade ($y \in [1,7]$)			Approval rate ($y \in [0,1]$)		
k	Lower [CI]	Upper [CI]	Mid.	Lower [CI]	Upper [CI]	Mid.
$k = 2$	-2.742 $[-2.86, -2.66]$	3.506 $[3.42, 3.58]$	0.382	-0.327 $[-0.36, -0.30]$	0.715 $[0.69, 0.73]$	0.194
$k = 8$	-2.658 $[-2.74, -2.56]$	3.590 $[3.51, 3.69]$	0.466	-0.341 $[-0.36, -0.32]$	0.700 $[0.68, 0.73]$	0.179
$k = 10$	-2.694 $[-2.80, -2.59]$	3.554 $[3.48, 3.65]$	0.430	-0.369 $[-0.40, -0.34]$	0.672 $[0.65, 0.70]$	0.152

Notes: All bounds computed under the Manski (no-assumption) model. $k = 2$ maximizes the pass-rate midpoint ATE; $k = 8$ maximizes the grade midpoint ATE and is selected in 60.83% of bootstrap resamples. $k = 10$ is the program's official "active engagement" threshold per the Official Leveling Program guidelines. Bootstrap CIs are narrow (≤ 0.2 grade points and ≤ 0.04 pass-rate points), so the bound endpoints are estimated precisely; the dominant uncertainty is the assumption-driven width of the bound itself, not sampling noise.

Under the midpoint rule (choose k maximizing $(\tau^L + \tau^U)/2$), the data-driven recommendation is $k = 8$ for grades (midpoint 0.466, bootstrap-stable: selected in 60.83% of resamples) and $k = 2$ for approval rates (midpoint 0.194). However, the program guidelines define active engagement as attending at least 10 sessions. At $k = 10$, the grade midpoint remains 0.430 and the pass-rate midpoint is 0.152, both positive and close to the respective empirical optima. The official threshold is therefore a defensible operational choice, nearly as informative as the data-driven optimum.

Under MTR (Assumption 1), the grade interval at $k = 8$ collapses from $[-2.658, 3.590]$ to $[0, 3.590]$, giving a midpoint of 1.795, a substantial improvement, but only if one is willing to rule out any possibility that tutoring harms students.

5.3 CTE Sensitivity: Pass-vs-Fail Identification

The ATE bounds on grades remain wide because the outcome support $[1, 7]$ mechanically forces the Manski width to 6 grade points (Remark 1). A natural way to sharpen what the data can say is to switch from the ATE to a parameter that respects stochastic dominance (§3.3 of the lecture notes): the *Classical Treatment Effect* (CTE) at outcome threshold y^*,

$$\mathrm{CTE}_k(y^*) \;=\; P(Y_1 \leq y^*) - P(Y_0 \leq y^*). \tag{7}$$

Because the unidentified counterfactual probabilities $P(Y_1 \leq y^* \mid D = 0)$ and $P(Y_0 \leq y^* \mid D = 1)$ live in $[0, 1]$ rather than the $[1, 7]$ outcome range, the Manski-type bounds on the CTE are mechanically tighter than those on the ATE. A *negative* CTE at threshold y^* means the treated outcome distribution stochastically dominates the control distribution at that point, i.e. tutoring shifts probability mass above y^*.

Applying the same Law-of-Total-Probability decomposition used in Section 1, the Manski sharp bounds on the CTE are

$$\mathrm{CTE}^L_k(y^*) = F_1\, P_1 - (F_0\, P_0 + P_1), \tag{8}$$
$$\mathrm{CTE}^U_k(y^*) = F_1\, P_1 + P_0 - F_0\, P_0, \tag{9}$$

where $F_t = P(Y \leq y^* \mid Z_k = t)$ and $P_t = P(Z_k = t)$. Under MTR, the pointwise inequality $Y_1 \geq Y_0$ implies $P(Y_1 \leq y^*) \leq P(Y_0 \leq y^*)$, so the upper bound collapses to zero. Under distribution-MTS, the lower bound tightens to the naive estimate $F_1 - F_0$.

Figure 11 reports the CTE bounds across k for the passing-grade threshold $y^* = 4$ (BAP students, Semester 1). The Manski interval is still wide, but the picture sharpens: the MTS lower bound, equal to the naive CTE, stays uniformly negative around -0.1 to -0.2 across all k. Under MTS the data therefore identify a reduction of 10 to 22 percentage points in the probability of failing the passing grade. The MTR upper bound is zero by construction.

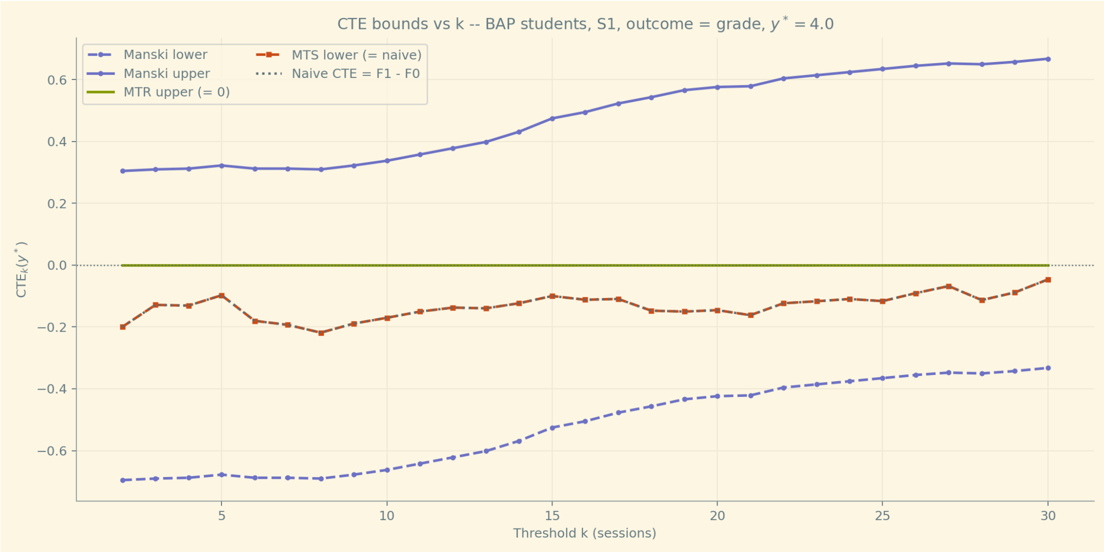

Figure 11: CTE bounds on the pass-vs-fail event $\{Y \leq 4\}$ as k varies, BAP students, Semester 1. The MTS lower bound (orange) is the naive CTE $F_1 - F_0$; the MTR upper bound (green) is identically zero. The Manski interval (violet) is wide for all k, but the MTS-tightened interval [naive, 0] identifies a robust negative CTE.

Extending this to a bivariate sweep, Figure 12 plots the naive CTE $F_1 - F_0$ over a grid of (k, y^*) values, with y^* covering the full grade range $[1.5, 6.5]$. The deepest negative values (darkest blue, ≈ -0.3) cluster around $y^* \in [3, 4.5]$ at moderate $k \in [4, 12]$: this is the pass-vs-fail region where tutoring most clearly shifts probability mass above the threshold. Above $y^* \approx 5.5$ the effect fades or reverses slightly, indicating that tutoring lifts strugglers up to the passing margin but does not push top performers further.

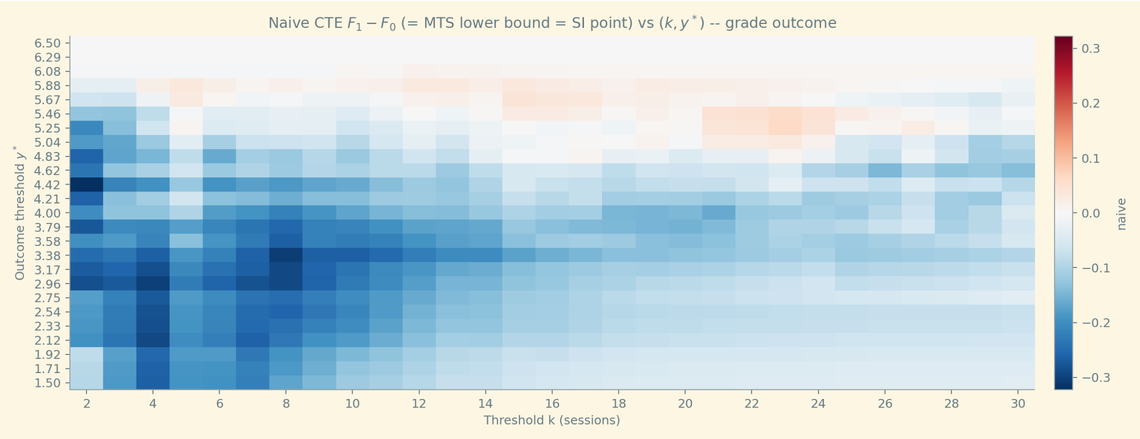

Figure 12: Naive CTE $F_1 - F_0$ over the bivariate grid (k, y^*) for the grade outcome. Blue cells indicate a negative CTE (treated dominates control at the threshold under MTS); the darkest blue region is the "pass-vs-fail" band $y^* \in [3, 4.5]$ at $k \in [4, 12]$.

The same heatmap for the pass-rate outcome (Figure 13) confirms the pattern: tutoring is most clearly beneficial in shifting mass above moderate pass-rate thresholds ($y^* \in [0.5, 0.8]$) at intermediate intensities.

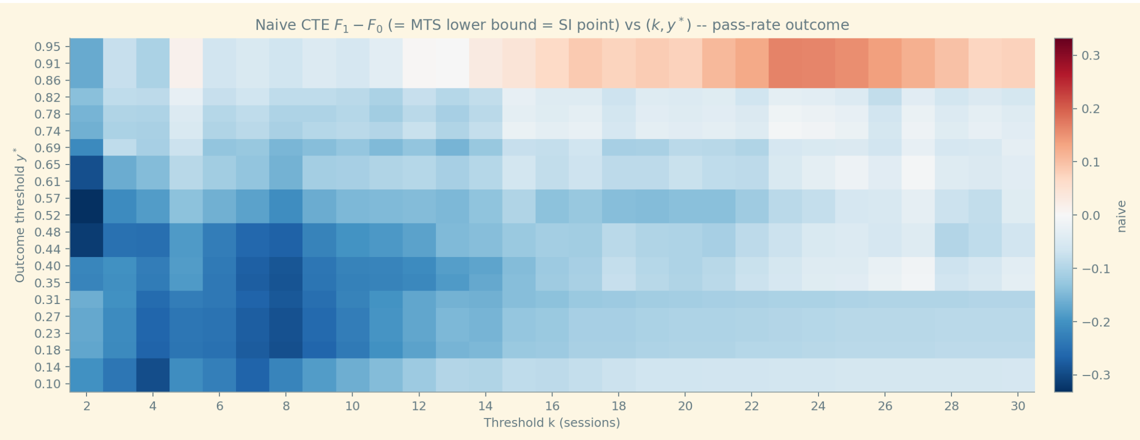

Figure 13: Naive CTE $F_1 - F_0$ over the bivariate grid (k, y^*) for the pass-rate outcome.

The CTE perspective therefore complements the ATE analysis: even where the ATE is unsigned under Manski, the CTE identifies (i) a target outcome region (the passing margin, $y^* \approx 4$) and (ii) a target intensity range ($k \in [4, 12]$) where tutoring is most clearly beneficial under the additional restriction that selection into tutoring respects stochastic dominance.

6 Continue vs. Opt-out?

In this section treatment is defined as continuing tutoring in semester 2. The analysis compares BAP and VAI students and evaluates both grades and approval rates. Before proceeding, it is useful to note how many students decided to continue or opt out. Figure 14 shows that approximately 75.7% of BAP students continued in the program.

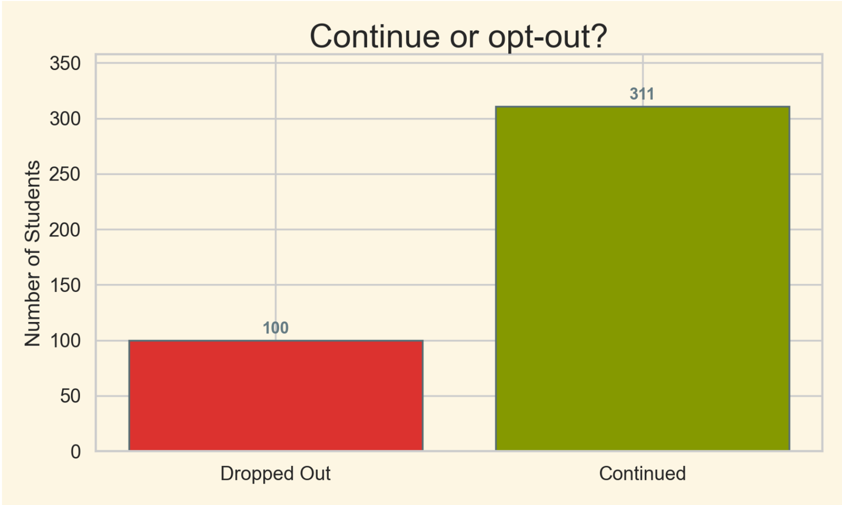

Figure 14: Number of BAP students who continued vs. opted out of the program between semesters 1 and 2

Table 2 reports the Manski bounds and minimax-regret treatment shares for all four subgroup-outcome combinations.

Table 2: Manski ATE bounds and minimax-regret shares for Continue vs. Opt-out (Semester 2). Bracketed values are 5th–95th percentile bootstrap CIs (120 resamples) on each endpoint.

		ATE bounds				
Group	Outcome	τ^L [CI]	τ^U [CI]	Mid.	Width	δ^*
BAP	Grade ($y \in [1, 7]$)	-3.395 $[-3.50, -3.26]$	3.743 $[3.63, 3.86]$	0.174	7.139	0.524
BAP	Pass-rate ($y \in [0, 1]$)	-0.479 $[-0.51, -0.45]$	0.710 $[0.68, 0.74]$	0.115	1.190	0.597
VAI	Grade ($y \in [1, 7]$)	-3.525 $[-3.66, -3.37]$	3.787 $[3.67, 3.91]$	0.131	7.312	0.518
VAI	Pass-rate ($y \in [0, 1]$)	-0.516 $[-0.55, -0.48]$	0.702 $[0.67, 0.73]$	0.093	1.219	0.577

Notes: $\delta^* = \tau^U/(\tau^U - \tau^L)$ is the minimax-regret optimal share that a planner would assign to Continue if continuation could be centrally re-assigned. The observed continuation rate among BAP students (75.7%) reflects self-selection and is not directly comparable to δ^*. All intervals contain zero; no sign of the ATE is robustly identified.

Figure 15 reports the partially identified ATE of continuing in the tutoring program during the second semester versus opting out, for both BAP and VAI students and for the two outcomes. In all four cases the estimated ATE interval includes zero, so the data do not allow us to conclude that either decision systematically dominates the other.

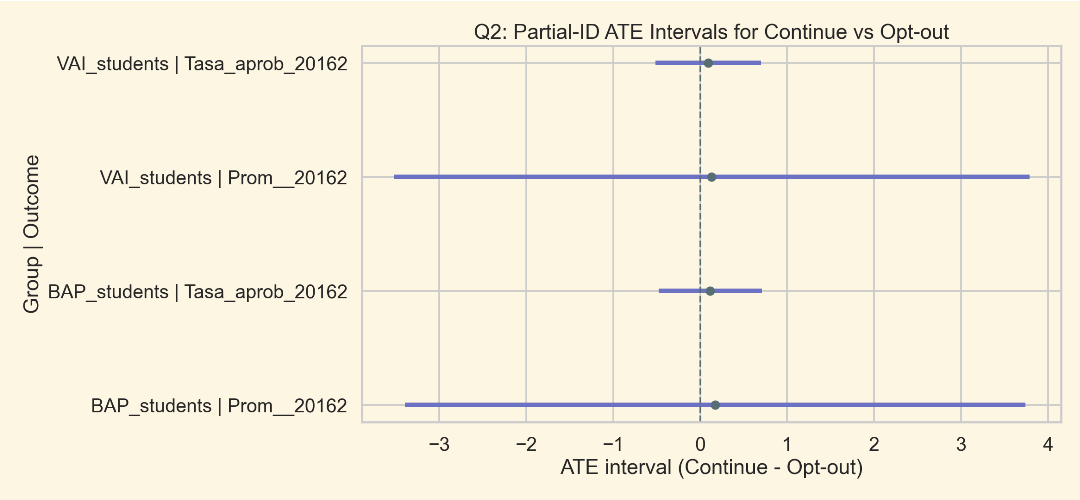

Figure 15: Q2 partial-identification intervals for Continue versus Opt-out.

The partial-ID intervals are ambiguous in sign for all subgroup-outcome combinations. The conservative and optimistic decision rules, as well as the minimax-regret criterion, are discussed in Section 7.1 below.

6.1 Assumption Ladder

We now add MTR and MTS tightening to further explore the robustness of the Q2 results.

Figure 16 replicates the previous figure under three sets of assumptions. The Manski, MTR, and MTS intervals are still fairly wide and in every case contain zero, so even with these structured restrictions the data remain consistent with small positive, null, or small negative effects of continuation on both grades and approval rates.

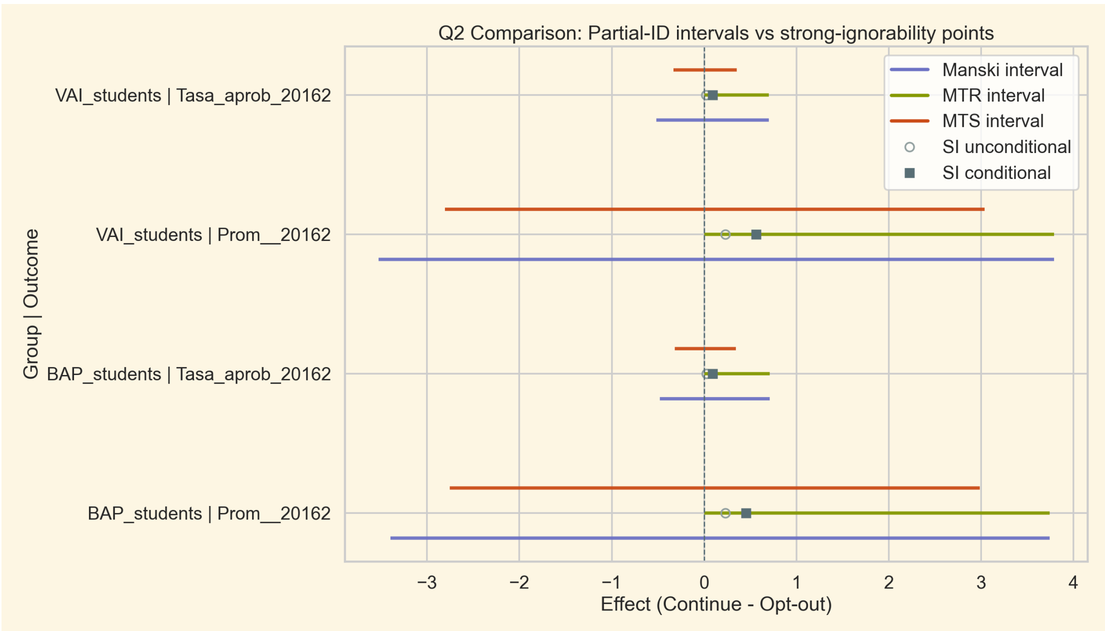

Figure 16: Q2: partial-identification intervals versus strong-ignorability point estimates. Open circles denote the unconditional strong-ignorability estimate; filled squares denote the conditional (covariate-adjusted) version.

Figure 17 makes the interval narrowing quantitative. Under Manski the median ATE width is approximately 4 grade points; MTR roughly halves it to a median of ≈ 2; MTS provides an intermediate reduction. Crucially, even under MTR the spread remains substantial, so no assumption in the ladder produces a tight enough interval to resolve the Continue/Opt-out decision unambiguously.

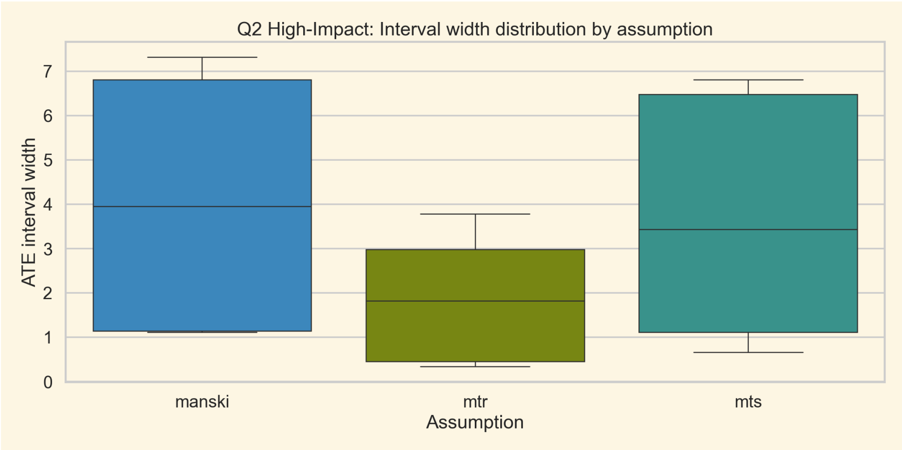

Figure 17: Distribution of ATE interval widths by assumption for Q2 (pooled over all subgroup-outcome combinations). MTR reduces median width from ≈ 4 to ≈ 2; MTS provides a weaker but still meaningful reduction. No assumption eliminates ambiguity in sign.

7 Discussion and Recommendations

7.1 Policymaker Decision Framework

Chapter 10 of the lecture notes presents four policymaker perspectives derived for a binary outcome $Y \in \{0, 1\}$, with interval expressions built from $P(Y = 1 \mid X)$ and $P(Y = 0, Z = z \mid X)$ (cf. the summary table in §10.2.6 of the notes). Those expressions translate directly to the pass-rate outcome $Y = \texttt{Tasa\_aprob\_20162} \in [0, 1]$, which is a probability proxy. For the grade outcome $Y \in [1, 7]$ the binary-outcome formulas do not have a natural counterpart, so the formal CPO/CPC/FBC/CBF analysis below is restricted to the pass-rate; for grades, only the data-derived decision rules (optimistic, conservative, minimax-regret) of §7.2 apply.

Let $P_1 = \Pr(\text{Continue} = 1)$, $P_0 = 1 - P_1$, $m_1 = \mathbb{E}[Y \mid \text{Continue} = 1]$, and $m_0 = \mathbb{E}[Y \mid \text{Continue} = 0]$, all conditional on observing Y. Applying the binary-outcome formulas of §10.2 to a $[0, 1]$-bounded outcome, the perspective-specific bounds on the ATE $\tau = \mathbb{E}[Y(1) - Y(0)]$ become:

- **CPO** (continuation weakly benefits everyone): $\tau \in [0,\ m_1 P_1 + P_0 - m_0 P_0]$.
- **CPC** (continuation weakly harms everyone): $\tau \in [m_1 P_1 - P_1 - m_0 P_0,\ 0]$.
- **FBC** (each arm is better off staying put): $\tau \in [-m_0 P_0,\ m_1 P_1]$.
- **CBF** (each arm would prefer switching): $\tau \in [-P_1(1 - m_1),\ P_0(1 - m_0)]$.

Table 3 reports the four intervals computed on the observed-outcome subsample for BAP and VAI.

Table 3: Perspective-specific ATE intervals for the pass-rate outcome (Continue vs. Opt-out, Semester 2). Computed on the observed-outcome subsample.

	BAP ($n = 333$, $P_1 = 0.88$)		VAI ($n = 293$, $P_1 = 0.85$)	
Perspective	τ^L	τ^U	τ^L	τ^U
Manski	−0.358	0.643	−0.381	0.619
CPO	0.000	0.643	0.000	0.619
CPC	−0.358	0.000	−0.381	0.000
FBC	−0.077	0.602	−0.099	0.564
CBF	−0.281	0.040	−0.282	0.054

Three observations follow. (i) CPO and CPC each cut the Manski interval at zero from opposite sides; both are consistent with the data but neither is empirically distinguishable from the other. (ii) FBC is markedly tighter than CBF on the upper side: the partial-ID interval allows substantial room for an upper bound near +0.6, but only if one is willing to assert that the factual arms outperform their counterfactuals. (iii) For both subgroups all four perspective intervals contain zero, so no perspective produces a robustly signed ATE; the action recommended by each perspective is the one Chapter 10 prescribes (CPO and CBF: continue; CPC and FBC: opt-out for whichever arm has the larger gap), with no data-driven preference between perspectives.

7.2 Decision Rules Applied to the Data

Given that all four Q2 ATE intervals contain zero, the three canonical rules give different answers.

Conservative rule (maximin): choose Continue only if $\tau^L \geq 0$. In all four subgroup-outcome combinations $\tau^L < 0$, so the conservative rule recommends *Opt-out*. Interpretation: a policymaker unwilling to risk any harm to any student should not mandate continuation.

Optimistic rule: choose the action with the better upper bound. For all four cases $|\tau^U| > |\tau^L|$ (BAP grades: $3.74 > 3.40$; BAP pass-rate: $0.71 > 0.48$; VAI grades: $3.79 > 3.52$; VAI pass-rate: $0.70 > 0.52$), so the optimistic rule recommends *Continue*.

Minimax-regret rule: this rule minimizes the worst-case opportunity cost across all data-compatible parameter values. For an interval $[\tau^L, \tau^U]$, the minimax-regret treatment share is

$$\delta^* = \frac{\tau^U}{\tau^U - \tau^L}.$$

For BAP grades: $\delta^* = 3.743/(3.743 + 3.395) \approx 0.524$. The interpretation is prescriptive, not descriptive: if the program could re-assign continuation centrally, minimax regret would assign approximately 52% of BAP students to Continue. The observed 75.7% continuation rate is the result of student self-selection, not of planner assignment, so it is not directly comparable to δ^*. The gap simply records that a minimax-regret planner would, in expectation, send fewer students to the program than choose to enrol on their own.

7.3 Main Recommendations

1. **Primary causal claim**: the Manski bounds are the main causal answer, since they require only bounded support they are the most credible and nearly universal assumption.

Under this standard, no sign of the ATE is robustly established for any subgroup or outcome in either semester.

2. **Threshold for Q1**: use $k = 10$ (the program's operational "active engagement" definition) as the primary reporting threshold. The grade midpoint at $k = 10$ is 0.430 and the pass-rate midpoint is 0.152, both positive but not robustly so. The data-driven optimum $k = 8$ (midpoint 0.466, bootstrap-stable) is close and could serve as an alternative if program guidelines are updated.
3. **Q2 policy**: the optimistic policymaker recommends Continue; the conservative (maximin) policymaker recommends Opt-out. They reflect different attitudes toward worst-case risk. The honest econometrician should present both and let the policymaker declare their risk posture.
4. **Strong-ignorability as benchmark**: the unconditional Q2 estimates are +0.231 grade ATE and +0.026 pass-rate ATE for BAP, and +0.230 and +0.021 for VAI. Adjusting for baseline covariates (PTJE_RANKING, PTJE_LYC, PTJE_MAT, and the complementary group indicator) gives substantially larger estimates: +0.452 grade ATE and +0.092 pass-rate ATE for BAP, and +0.562 and +0.094 for VAI. The two versions bracket plausible point identifications: the unconditional one is implausibly contaminated by negative selection into tutoring (low-baseline students are over-represented among continuers), while the conditional one removes the part of that selection captured by the observed covariates. Both estimates sit well inside the Manski intervals, and the conditional one approaches the MTS upper region for grades.
5. **Use MTR/MTS as robustness checks**: MTR halves the grade interval width at the cost of ruling out harm; MTS provides a middle ground. Both should be presented as sensitivity analyses, not as the main result.

8 Conclusion

This paper applied the partial identification framework to evaluate tutoring effects in a Chilean university leveling program. The main findings are:

(i) Under bounded support only (Manski bounds), the sign of the ATE is not established for any threshold k or any subgroup in either semester.

(ii) Adding MTR (Assumption 1) forces the grade lower bound to zero and halves the interval width at $k = 8$, but this result requires ruling out any possibility that tutoring harms students.

(iii) The program's official threshold $k = 10$ yields a grade midpoint of 0.430, close to the empirically optimal $k = 8$ (0.466), making it a defensible operational choice.

(iv) Under strong ignorability (Assumption 3), the unconditional point estimates are small and positive ($\approx$0.23 grade points for both BAP and VAI); the conditional version, which adjusts for baseline admission scores and the complementary group indicator, is roughly twice as large (+0.45 for BAP grades and +0.56 for VAI grades). The conditional version is more defensible than the unconditional one, but still relies on the untestable assumption that the four observed covariates exhaust the relevant sources of selection.

(v) For Q2, the conservative (maximin) rule recommends Opt-out; the optimistic rule recommends Continue; under minimax regret, a planner with re-assignment authority would send $\approx 52\%$ of BAP students to Continue, fewer than the 75.7% who currently self-select into the program.

These results illustrate Manski's Law of Decreasing Credibility: stronger assumptions yield sharper conclusions but at the cost of credibility. The appropriate policy response depends on the policymaker's risk posture, which can be formalized through the CPO/CPC/FBC/CBF framework and minimax-regret decision rules.

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